MILITARY SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE), ESTABLISHED RELIABILITY AND NON-ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-C-39018B, dated 23 March 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

3.5.2., line 4: Delete "70" and substitute "100".

PAGE 6

3.31 lines 5, 6, and 7: Delete "Polarity on polarized capacitors with axial wire-lead terminals . . . case." and substitute "Polarity on polarized capacitors with axial wire-lead terminals shall be indicated by a minimum of two plus (+) symbols at the positive end of the case."

PAGE 11

TABLE III, Subgroup 1: Following terminal strength, add "(Not applicable to style CU81)".

PAGE 12

4.5.2.1.1: Delete and substitute:

"4.5.2.1.1 Sampling plan . Every 2 months (every year for styles CU14, CU15, and CU81), the inspections shall consist of subgroups 1, 2, and 3. Every 6 months (every 2 years for styles CU14, CU15, and CU81), the inspections shall consist of subgroups 4 through 9. Sampling shall be as follows:

a. Every 2 months (every year for styles CU14, CU15, and CU81), sample units from production including those with the largest capacitance voltage product per case size shall be selected at random from each style in production that has passed group A inspection.
b. Every 6 months (every 2 years for styles CU14, CU15, and CU81), sample units to be subjected to the tests of subgroups 4, 5, 7, 8, and 9 shall be selected at random from units that have passed the group A inspection. Sample units to be subjected to the test of subgroup 6 shall be randomly selected every 6 months (every 2 years for styles CU14, CU15, and CU81), including units of the largest capacitance value in each style (insulated or uninsulated), and voltage group that have passed group A inspection and subgroup 2 of the group B inspection.
c. When the samples are selected, the contractor shall review all selections made within the preceeding time period in order to assure that all styles in production have been drawn into the test program."

PAGE 13

TABLE IV, subgroup 3, columns 5 and 6: Delete "0" failures allowed and substitute "1" failure allowed.

TABLE IV, column 6: Delete the bracket after subgroup 3.

TABLE IV, footnote 2/: Delete footnote 2/ and reference to it in column 5.

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TABLE IV, subgroup 4: Add:

"Terminal strength (CU81 only) -	3.12	4.6.9	12	1	NA "

TABLE IV, subgroup 6, Life (10,000 hrs) (ER only), column 6; Delete "1" failure allowed and substitute "4/".

PAGE 14

4.6.3, first sentence: Delete and substitute: "Capacitors shall be subjected to 100 percent of the dc rated voltage at the high ambient test temperature for 48 hours minimum but not to exceed 96 hours".

PAGE 15

4.6.10.1: Delete and substitute:

"4.6.10.1 Impedance (see 3.13). Impedance shall be measured at -55°C, +0°C, -3°C (or as specified, see 3.1) in accordance with 4.6.1.1. The following current limits shall be used when measuring impedance:

Max current level (mA)	Range of expected impedance (ohms)		
100	.001	to	.1
10	.03	to	3
1	1.0	to	30
.1	10	to	300
.01	100	to	3000

These values are sufficiently low to prevent a detectable change in impedance due to internal heating of the capacitor. Measurement accuracy shall be within ±5 percent."

4.6.11.1c, second sentence: Delete and substitute: "For capacitors covered by MIL-C-39018/1 and MIL-C-39018/2, a quantity of capacitors equal to the quantity tested at 85°C shall be tested at 125°C".

PAGE 19

4.6.26, line 7: Delete "6.6" and substitute "6.5".

PAGE 20

Add the following new paragraph:

"6.8 Substitutability for lower vibration levels. Parts qualified and marked to high frequency vibration levels are substitutable for low frequency vibration level parts."

PAGE 23

TABLE VII, following entry for M39018/03-1098M, add:

"M39018/03-1209M M39018/03-1299M	66	Style CUR17 (high frequency vibration). All capacitance values in all voltage ratings of voltage group I.
M39018/03-1251M M39018/03-1283M	66	Style CUR17 (high frequency vibration). All capacitance values in all voltage ratings of voltage group II.
M39018/03-1291M M39018/03-1298M	66	Style CUR17 (high frequency vibration). All capacitance values in all voltage ratings of voltage group III."

TABLE VII, following entry for M39018/07-0128M, add:

"M39018/07-0212M M39018/07-0295M	63	Style CUR19 (high frequency vibration). All capacitance values in all voltage ratings of voltage group I.

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M39018/07-0304M	63	Style CUR19 (high frequency vibration). All capacitance
M39018/07-0328M	63	values in all voltage ratings of voltage group II."
M39018/08-0001M	51	Style CU01. All capacitance values in all voltage
M39018/08-0012M	51	ratings of voltage group I.
M39018/08-0013M	51	Style CU01. All capacitance values in all voltage
M39018/08-0018M	51	ratings of voltage group II.
M39018/08-0102M	63	Style CUR01. All capacitance values in all voltage
M39018/08-0113M	63	ratings of voltage group I.
M39018/08-0114M	63	Style CUR01. All capacitance values in all voltage
M39018/08-0119M	63	ratings of voltage group II.
M39018/09-0018M	63	Style CUR02. All capacitance values in all voltage
M39018/09-0077M	63	ratings of voltage group I.
M39018/09-0087M	63	Style CUR02. All capacitance values in all voltage
M39018/09-0176M	63	ratings of voltage group II.
* M39018/10-0001M	27	Style CUR92. All capacitance values in all voltage
* M39018/10-0011M	27	ratings of voltage group I, terminal style A.
* M39018/10-0141M	27	
* M39018/10-0151M	27	
* M39018/10-0012M	27	Style CUR92. All capacitance values in all voltage
* M39018/10-0020M	27	ratings of voltage group I, terminal style B.
* M39018/10-0152M	27	
* M39018/10-0160M	27	
* M39018/10-0161M	27	Style CUR92. All capacitance values in all voltage
* M39018/10-0171M	27	ratings of voltage group II, terminal style A.
* M39018/10-0201M	27	
* M39018/10-0211M	27	
* M39018/10-0172M	27	Style CUR92. All capacitance values in all voltage
* M39018/10-0180M	27	ratings of voltage group II, terminal style B.
* M39018/10-0212M	27	
* M39018/10-0220M	27	

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